A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 11

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Radiation absorbed by tissue produces free radicals (ions with unpaired electrons), mostly from water, which are highly reactive **1** Free radicals damage DNA / chromosomes in the nucleus causing mutations (cancer, inherited effects) - rapidly dividing cells (skin, gut, bone marrow) most affected **1** Damage to other cell parts can stop the cell functioning or kill it – low damage may be repaired by the intact nucleus **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	3
01.2	**1 mark each** 1 mark for each of the 2 points below. **1 Acute:** effects appearing quickly after a large dose over a short time (radiation sickness, damage to bone marrow / immune system / gut, haemorrhage, rapid death) **1 Chronic:** effects appearing slowly, in proportion to the dose received over a long period (cancers, cataracts) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1 Exposure:** absorption of ionising radiation from a source the person is close enough to **1 Contamination:** physically carrying radioactive material on or in the body, so exposure continues wherever the person goes – sealed sources can expose but not contaminate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	2
01.4	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Sieverts = grays x radiation weighting factor (20 for alpha) **1** 0.002 x 20 = 0.04 Sv (40 mSv) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (Sv) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
01.5	1 mark for the point below. **1** The activity of a source (number of decays per second) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **becquerel** → one radioactive decay per second	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 301, 303; AQA spec 3.4.3.2.14.	
02.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (150 - 125) / 150 x 100 = 16.7% (accept 17%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** A fixed fee gives no incentive to save water because the bill is the same however much is used **1** Metering matches the charge to the volume used so households save money by using less (fewer hosepipes, low-flow appliances) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Low-water appliances:** dual-flush toilets, low-flow shower heads, efficient washing machines / dishwashers, tap timers **1 Greywater re-use:** bath / shower water used to flush toilets or water gardens **1** Rainwater collection / water butts for garden use **1 Xeriscaping:** planting drought-adapted species that need no watering **1** Repairing leaks / dripping taps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Old infrastructure / pipes in poor condition leak **1** Repair and good maintenance / pipe replacement / leak detection reduce wastage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.5	**1 mark each** 1 mark for each of the 2 points below. **1** Water is delivered directly to the plant roots / soil around each plant **1** Less evaporation loss from spray droplets and from wetting the whole soil surface – less runoff **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 124-126; AQA spec 3.2.2.5.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Organic (methyl) mercury **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Anaerobic microbes converted the inorganic mercury into organic methyl mercury **1** Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most heavy metals are less soluble at high pH **1** So they remain immobile and are not leached into water or absorbed by organisms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 5 points below. **1** Use a calibrated sound meter (dB(A)) at fixed distances (eg 10, 50, 100, 250, 500 m) from the track behind each barrier and at a control site with no barrier **1 Standardise the noise source:** same cars / same race or a standard test noise – same time of day, wind speed / direction, weather – same height of meter above ground **1** Take repeated readings at each distance and calculate means (and standard deviation) to assess variability **1** Use a measure such as LA10 or Leq over a set period rather than a single peak reading **1** Compare reduction in dB relative to the control at each distance – plot noise level against distance for each barrier – statistical test (t-test) to assess significance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **standard deviation** → a measure of the spread of the data about the mean • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	5
04.2	1 mark for the point below. **1** Wind carries sound / affects how far it travels so different wind directions would change readings independently of the barrier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** LA10 is the noise level exceeded for 10% of the period - a measure of the noisiest periods **1** LA90 is the level exceeded for 90% of the period - a measure of the quietest / background level **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Double / triple glazing **1** Acoustic insulation in walls / roof **1** Bedrooms on the side facing away from the track **1** Dense hedges / trees and earth mounds in gardens **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 293-295, 418; AQA spec 3.4.3.2.13.	
05.1	**1 mark each** 1 mark for each of the 4 points below. **1 Gravimetry:** gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) **1** Magnetometry **1 Seismic survey:** sound / shock waves from controlled explosions or a seismic vibrator – echoes recorded by geophones give the depth, density and shape of rock strata **1** Resistivity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Trial drilling is the most expensive technique per site **1** Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites **1** Drilling is then targeted where deposits are likely – it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Different minerals emit / reflect infrared at different wavelengths **1** Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	1 mark for the point below. **1** Results are available immediately in the field, saving time waiting for laboratory analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 137; AQA spec 3.2.3.4.	
06.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 7×10^{11} tonnes **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
06.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 10% of 10 000 = 1000 billion tonnes **1** 1000 / 700 x 100 = 143% (accept 142.9) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Carbon is transferred into and out of rocks extremely slowly (fossilisation / sedimentation, weathering, volcanic release) **1** Residence time is millions of years so the flux to the atmosphere is tiny compared with respiration / photosynthesis / combustion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
06.4	**1 mark each** 1 mark for each of the 3 points below. **1** CO_2 dissolves in the sea (forming carbonic acid / hydrogen carbonate ions) and is absorbed by phytoplankton through photosynthesis **1** Organisms such as molluscs, corals and planktonic organisms build calcium carbonate shells / exoskeletons **1** When they die the shells sink and sediment on the seabed, becoming limestone / chalk over long periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
06.5	**1 mark each** 1 mark for each of the 2 points below. **1** Aerobic respiration uses oxygen to break down organic compounds completely to CO_2 (and water), releasing much more energy **1** Without oxygen, anaerobic organisms only partly break down organic compounds, releasing carbon as methane (CH_4), which is later oxidised to CO_2 in the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-149; AQA spec 3.2.4.2.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Surplus energy drives a pump compressing air stored in underground caverns (salt mines) – releasing it drives machinery / turbines **1** Compression produces a lot of heat – storing this heat (hot oil / molten salt) and returning it when the air expands recovers energy that would otherwise be lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** As volume increases the surface area : volume ratio decreases **1** Heat is lost through the surface so there is less surface per unit of stored heat – very large stores can carry summer solar heat into winter (inter-seasonal storage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Potassium nitrate has a much higher boiling point so it stores heat at up to 550 °C without pressurisation **1** Water could only reach such temperatures under high pressure, increasing costs – high temperature needed to raise steam for turbines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Materials with high specific heat capacity (concrete, water) warm and cool slowly **1** They absorb heat in hot weather reducing overheating / air conditioning, and release it as the weather cools, delaying the need for heating **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.5	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Flywheels:** rotating mass stores kinetic energy for short-term smoothing / generation **1 Supercapacitors:** electrochemical capacitors with 10-100 times the energy density of conventional capacitors – possible future large-scale electricity storage **1** Thermal storage in molten salt / high-volume tanks – hydrogen / P2G – batteries **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222-224; AQA spec 3.3.4.1.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable **1** Wild species have different ranges of tolerance so cress results cannot be applied directly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated • **tolerance** → the range of conditions in which a species can survive **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Preliminary study with widely spaced pH values to find where germination occurs **1** Final range from below the lowest pH giving germination to above the highest, with closer intervals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.3	**1 mark each** 1 mark for each of the 4 points below. **1** Temperature (constant, natural germination temperature) **1** Light, water volume, number of seeds per dish, seed source **1** Growth medium (filter paper; soil would alter pH) – nutrients not needed for germination **1** Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surface water affects fresh mass and drying damages the plants – dry mass requires killing the plants so the same plants cannot be re-measured **1** Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.	
09.1	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Overgrazing:** vegetation is eaten faster than it grows so soil is exposed – hooves disturb the soil, damage roots and compact it, reducing infiltration and increasing runoff **2** Reduced biota (ploughing, agrochemicals, less organic matter): less humus so less adhesion between particles – fewer worms so less aeration and lower infiltration capacity – fewer nutrients released so less vegetation cover **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface • **humus** → dark, decomposed organic matter in soil **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Tree roots that held the soil together decompose so the soil is bound less strongly **1** Heavy rain adds mass and lubricates particles – without interception more water enters the soil, making slumping more likely **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	2
09.3	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Sedimentation in rivers reduces flow capacity so flooding downstream is more likely **1** Sedimentation in reservoirs reduces their storage volume **1** Coastal sedimentation covering reefs / seagrass – turbidity reducing light for aquatic plants **1** Increased atmospheric particulates from wind erosion worsening asthma – dust covering crops / urban areas **1** Desertification as vegetation cannot re-establish – reduced rainfall **1** Nutrients (phosphates) carried on soil particles causing eutrophication **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **particulates** → fine solid particles / smoke and dust • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Neonicotinoids are neurotoxins similar to nicotine – they inhibit the neurotransmitter acetylcholine • High toxicity to insects but low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier • **Very toxic to bees:** sub-lethal doses reduce navigation ability, foraging and immunity to disease – linked to colony decline • **Water soluble and moderately persistent:** remain in soil, absorbed by roots and translocated (systemic), so present in pollen and nectar • Persistence may allow build-up in aquifers / soil – broken down fastest in sunlight and by soil microbes • Synergism with some fungicides increases toxicity to bees • Exposure greatest when sprayed onto crops or when seed-coating dust is blown by wind • **Control:** seed coating rather than spraying – spraying when crops are not in flower / at night – still days to reduce drift • Restrictions on use (EU bans on outdoor use on flowering crops) – use of alternative pest control (cultural, biological, integrated control) • **Evaluation:** reduced pesticide use but neonicotinoid alternatives (pyrethroids) also harm pollinators – trade-off with crop yields **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **persistent** → does not break down / stays in the environment for a long time • **Synergism** → two pollutants together doing more damage than the sum of each alone **Do not credit**	9
	• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Neonicotinoids are neurotoxins related to nicotine that block the neurotransmitter acetylcholine. They are highly toxic to insects but of low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier - the specificity that made them popular. Bees, however, are insects: even sub-lethal doses impair navigation, foraging and immunity, and this has been linked to colony decline. Because the compounds are water-soluble and moderately persistent they remain in the soil, are absorbed by roots and translocated through the plant (systemic action), so they appear in pollen and nectar that bees collect; persistence may also allow build-up in soils and aquifers, although sunlight and soil microbes break them down. Some fungicides act synergistically with them, increasing toxicity to bees.	
	Exposure is greatest when crops are sprayed or when dust from coated seeds is blown by wind, so impacts are reduced by using seed coatings instead of sprays, spraying only when crops are not in flower or at night when bees are not foraging, and applying on still days to limit drift. The EU has banned outdoor use on flowering crops, and integrated control with cultural and biological methods reduces the need for insecticides altogether.	
	These measures reduce bee exposure, but the alternatives farmers turn to, such as pyrethroids, also harm pollinators, and restrictions can reduce yields, so the trade-off between crop protection and pollination services remains.	
	Weaker answer - Level 2 (4 marks)	
	Neonicotinoids are insecticides that kill insects by affecting their nervous system. They are systemic so they get into the whole plant including the pollen, and bees pick them up when they collect pollen and nectar. This harms bees and can cause colonies to die. To reduce the impact, farmers can spray when crops are not flowering, and some neonicotinoids have been banned in the EU. Other methods of pest control could be used instead.	
	What makes the difference	
	• Level 2 links the systemic property to bee exposure correctly but omits the other properties (solubility, persistence, specificity via the blood-brain barrier, synergism) that the question's 'properties' demands.	
	• Level 3 explains the sub-lethal effects (navigation, foraging, immunity) rather than 'harms bees'.	
	• Level 3 includes the trade-off evaluation (alternatives also harm pollinators, yield costs) that the mark scheme rewards.	

Do not write outside the box

Question	Marking guidance	Marks
10.2	**Any 1 of 6** 1 mark each for any 1 of the 6 points below, in any order, maximum 1. **1** Biological control (introduced predators / pathogens) **1** Predator habitats (hedgerows, beetle banks) **1** Crop rotation **1** Companion / barrier crops **1** Sterile male technique / pheromone traps **1** Genetic resistance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 265-266; AQA spec 3.4.3.2.8.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Hydrological cycle processes and reservoirs – residence times – dynamic equilibrium – increasing demand (population, affluence, irrigation, industry) • **Human impacts on transfer rates:** deforestation, agriculture, urbanisation, climate change, abstraction – consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • **Case studies of unsustainable exploitation:** Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • **Sustainable strategies:** metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair – new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis – pollution control (sewage, AMD, buffer strips)	25
	• Links to thermohaline circulation and climate – energy costs – circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	

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Question	Marking guidance	Marks
11.2	**Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Why non-renewable:** formed too slowly – geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock – Lasky's principle – cut-off ore grade and its drivers • **Extraction technologies:** mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • **Reducing demand:** recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • **Environmental limits:** impacts of mining and their control – energy costs rising with lower grades – conflicts	25
	• **Judgement:** technology can extend supplies substantially but only within energy and environmental constraints – circular economy essential *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	

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